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Thermodynamic assessments of the Ag–Gd and Ag–Nd systems

S.L. Wang[a,b], C.P. Wang[a], X.J. Liu[a,*], K. Ishida[c]

[a] Department of Materials Science and Engineering, College of Materials, and Research Center of Materials Design and Applications, Xiamen University, Xiamen 361005, PR China
[b] College of Chemistry and Chemical Engineering, Xiamen University, Xiamen 361005, PR China
[c] Department of Materials Science, Graduate School of Engineering, Tohoku University, Sendai 980-8579, Japan

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ABSTRACT

The phase diagrams and thermodynamic properties in the Ag–Re (Re: Gd, Nd) binary systems have been assessed by using the CALPHAD (Calculation of Phase Diagrams) method on the basis of the experimental data including the thermodynamic properties and phase equilibria. The Gibbs free energies of the liquid, bcc, fcc, dhcp and hcp phases were described by the subregular solution model with the Redlich–Kister equation, and those of the intermetallic compounds ($Ag_{51}Gd_{14}$, Ag_2Gd, AgGd, $Ag_{51}Nd_{14}$, αAg_2Nd, βAg_2Nd and AgNd phases) in these two binary systems were described by the sublattice model. The thermodynamic parameters of each phase in the Ag–Re (Re: Gd, Nd) binary systems were obtained, and an agreement between the calculated results and experimental data was obtained in each binary system.

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1. Introduction

The intermetallic compounds consisted of rare earth (Re) elements and transition metals are of particular interest regarding their outstanding magnetic properties [1] and their reversible absorption of hydrogen gas at room temperature and nearly atmospheric pressure [2]. Ag–Re binary alloys are potential electronic materials and amorphous materials [3,4]. Therefore, it is important to understand the phase equilibria in the Ag–Re alloy systems. In the present paper, the thermodynamic descriptions for the phase diagrams in the Ag–Re (Re: Gd, Nd) binary systems were carried out by means of the CALPHAD (Calculation of Phase Diagrams) method. In this method, the thermodynamic properties of the Ag–Re (Re: Gd, Nd) alloy systems are studied using thermodynamic models. The thermodynamic parameters to describe the Gibbs free energy of the individual phases are optimized based on the experimental data on thermodynamic properties and phase diagram.

2. Experimental information

2.1. The Ag–Gd system

The Ag–Gd system is consisted of three solid solution phases (fcc (Ag), αGd and βGd phases) and three intermetallic compounds ($Ag_{51}Gd_{14}$, Ag_2Gd and AgGd phases). Based on the thermal analysis, microscopical examination and X-ray diffraction, Kiessler et al. [5] firstly investigated the phase diagram in the Ag–Gd system and reported three intermetallic compounds (Ag_3Gd, Ag_2Gd and AgGd phases) and the peritectic reaction (L (35 at.% Gd) + $Ag_3Gd \leftrightarrow Ag_2Gd$ at 830 °C). Kiessler et al. [5] also reported the maximum solid solubility of Gd in the fcc (Ag) solution phase to be less than 2 at.% and determined three eutectic reactions: L (11 at.% Gd) ↔ (Ag) + Ag_3Gd at 780 °C, L (37 at.% Gd) ↔ Ag_2Gd + AgGd at 825 °C, and L (77.5 at.% Gd) ↔ AgGd + αGd at 790 °C. The allotropic transformation of αGd ↔ βGd at 1235 °C was not indicated in Ref. [5].

Using the X-ray parametric method, Gschneidner et al. [6] determined the eutectic temperature (L ↔ (Ag) + $Ag_{51}Gd_{14}$) to be 805 ± 3 °C and the terminal solid solubility in the fcc (Ag) solution phase to be 0.95 at.% Gd. Mcmasters et al. [7] reported that the stoichiometry of the Ag-rich phase to be $Ag_{51}Gd_{14}$ phase by X-ray powder diffraction method. Bailey and Kline [8] determined the crystal structure of the $Ag_{51}Gd_{14}$ phase. Therefore, the $Ag_{51}Gd_{14}$ phase is accepted in the present work.

The major differences between the reports of Gschneidner et al. [6] and Kiessler et al. [5] are the maximam solid solubility of Gd in the fcc (Ag) solution phase (0.95 or <2 at.% Gd) and the eutectic temperature (L ↔ (Ag) + $Ag_{51}Gd_{14}$ at 805 or 780 °C). The solid solubility and the eutectic temperature reported by Gschneidner et al. [6] were accepted in the Ag–Gd phase diagram compiled by Gschneidner and Calderwood [9] with some modifications from the experimental data reported by Kiessler et al. [5]. The phase diagram

* Corresponding author. Tel.: +86 592 2187888; fax: +86 592 2187966.
E-mail address: lxj@xmu.edu.cn (X.J. Liu).

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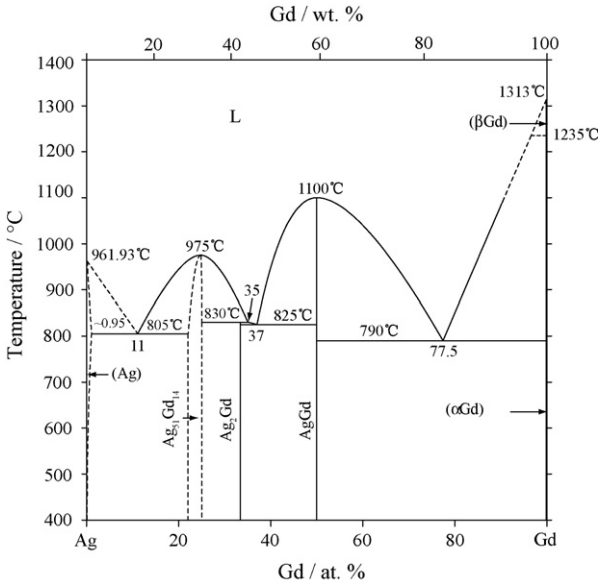

Fig. 1. The phase diagram of the Ag–Gd system reviewed by Gschneidner and Calderwood [9].

of the Ag–Gd system reviewed by Gschneidner and Calderwood [9] is shown in Fig. 1.

In addition, Fitzner and Kleppa [10] measured the standard enthalpies of formation of the $Ag_{51}Gd_{14}$ and AgGd phases, and the enthalpies of mixing of liquid phase at 1200 °C by high-temperature direct sysnthesis calorimetry, and Meschel and Kleppa [11] measured the standard enthalpy of formation of the Ag_2Gd phase. Batalin et al. [12] and Ivanov and Witusiewicz [13] reported the calculated mixing enthalpies of liquid phase at 1623 and 1600 K from somewhat scattered experimental values for the partial enthalpies measured by isoperibolic calorimetry. Ivanov and Lukashenko [14] measured activities of components and Gibbs free energies of formation of liquid phase at 1363 K by the Knudsen effusion method. The AgGd phase was found to be antiferromagnetic at low temperature by Sekizawa and Yasukochi [15], who reported that the Neel temperature of the AgGd phase is 145 K and Bohr magneton is 8.6.

Table 1
The stable solid phases and the used models in the Ag–Gd [9] and Ag–Nd [20] systems.

System	Phase	Structure report designation	Prototype	Modeling phase	Used models
Ag–Gd	(Ag)	A1	Cu	FCC	SSM
	$Ag_{51}Gd_{14}$	–	$Ag_{51}Gd_{14}$	$(Ag)_{51}(Gd)_{14}$	SM
	Ag_2Gd	C11	$MoSi_2$	$(Ag)_2(Gd)$	SM
	AgGd	B2	CsCl	(Ag)(Gd)	SM
	(αGd)	A3	Mg	HCP	SSM
	(βGd)	A2	W	BCC	SSM
Ag–Nd	(Ag)	A1	Cu	FCC	SSM
	$Ag_{51}Nd_{14}$	–	$Ag_{51}Gd_{14}$	$(Ag)_{51}(Nd)_{14}$	SM
	βAg_2Nd	C11	AlB_2	$(Ag)_2(Nd)$	SM
	αAg_2Nd	C11	$CeCu_2$	$(Ag)_2(Nd)$	SM
	AgNd	B2	CsCl	(Ag)(Nd)	SM
	(αNd)	A3	αLa	DHCP	SSM
	(βNd)	A2	W	BCC	SSM

Note: SSM—subregular solution model; SM—sublattice model.

2.2. The Ag–Nd system

The Ag–Nd system is consisted of three solid solution phases (fcc (Ag), αNd and βNd phases) and four intermetallic compounds ($Ag_{51}Nd_{14}$, αAg_2Nd, βAg_2Nd and AgNd phases). Based on chemical analysis, metallography, X-ray diffraction and DTA, the Ag–Nd system was firstly investigated by Ferro et al. [16], who observed three stoichiometry compound phases: $Ag_{51}Nd_{14}$ (melting point, 1025 ± 10 °C), Ag_2Nd (peritectic formation at 831 ± 5 °C) and AgNd (melting point, 960 ± 10 °C). Ferro et al. [16] also determined remelting reaction (βNd ↔ L + αNd at 725 °C) and three eutectic reactions: L (10.5 at.% Nd) ↔ (Ag) + $Ag_{51}Nd_{14}$ at 803 ± 5 °C, L (40.8 at.% Nd) ↔ Ag_2Nd + AgNd at 810 ± 5 °C, and L (79.5 at.% Nd) ↔ AgNd + αNd at 630 ± 5 °C. In the intermediate composition range, Ferro et al. [16] observed unexplained thermal effects at 783 ± 5 and 650 ± 10 °C. In the present work, the $Ag_{51}Nd_{14}$ (melting point, 1025 ± 10 °C) [16] was accepted.

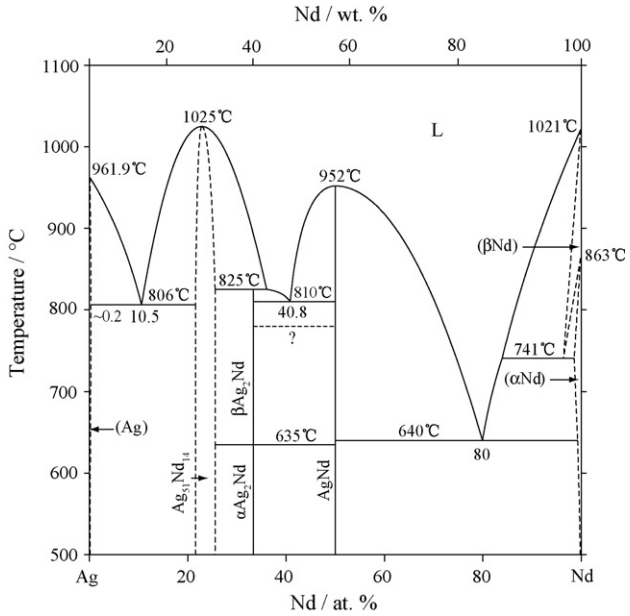

Fig. 2. The phase diagram of the Ag–Nd system reviewed by Gschneidner and Calderwood [20].

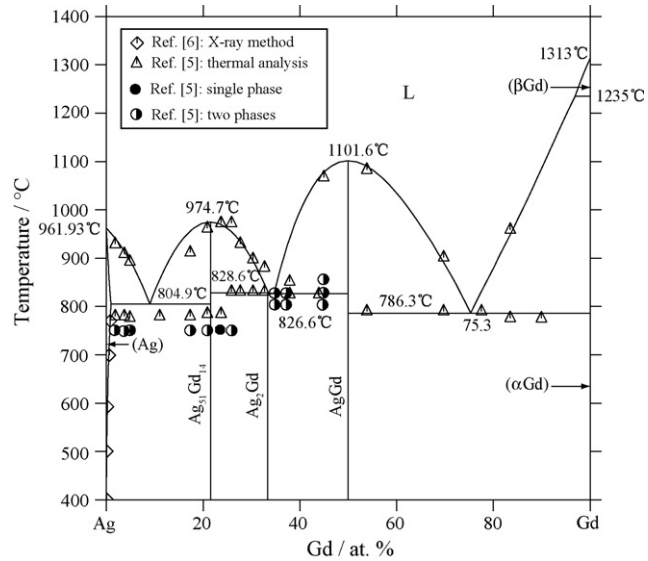

Fig. 3. The calculated phase diagram of the Ag–Gd system compared with the experimental data [5,6].

Table 2
Thermodynamic parameters in the Ag–Gd system assessed in the present work.

Parameters in each phase (J/mol)
Liquid phase, format (Ag, Gd)
${}^{0}L^{\mathrm{Liq}}_{\mathrm{Ag,Gd}} = -98000 + 51.934T - 3.946T\ln T$
${}^{1}L^{\mathrm{Liq}}_{\mathrm{Ag,Gd}} = -37550 + 4.271T$
${}^{3}L^{\mathrm{Liq}}_{\mathrm{Ag,Gd}} = 17892$
Fcc phase, format (Ag, Gd)
${}^{0}L^{\mathrm{Fcc}}_{\mathrm{Ag,Gd}} = -35458 + 7.017T$
${}^{1}L^{\mathrm{Fcc}}_{\mathrm{Ag,Gd}} = -24120$
Bcc phase, format (Ag, Gd)
${}^{0}L^{\mathrm{Bcc}}_{\mathrm{Ag,Gd}} = 20000$
Hcp phase, format (Ag, Gd)
${}^{0}L^{\mathrm{Hcp}}_{\mathrm{Ag,Gd}} = 30000$
$T_C^{\mathrm{Hcp}} = 293.4$
$\beta^{\mathrm{Hcp}} = 3$
AgGd phase, format $(\mathrm{Ag})_{0.5}(\mathrm{Gd})_{0.5}$
${}^{0}G^{\mathrm{AgGd}}_{\mathrm{Ag:Gd}} = 0.5 \times {}^{0}G^{\mathrm{fcc}}_{\mathrm{Ag}} + 0.5 \times {}^{0}G^{\mathrm{hcp}}_{\mathrm{Gd}} - 25000 + 0.135T$
$T_C^{\mathrm{AgGd}} = 145$
$\beta^{\mathrm{AgGd}} = 8.6$
$\mathrm{Ag_2Gd}$ phase, format $(\mathrm{Ag})_{0.667}(\mathrm{Gd})_{0.333}$
${}^{0}G^{\mathrm{Ag_2Gd}}_{\mathrm{Ag:Gd}} = 0.667 \times {}^{0}G^{\mathrm{fcc}}_{\mathrm{Ag}} + 0.333 \times {}^{0}G^{\mathrm{hcp}}_{\mathrm{Gd}} - 24900 + 2.503T$
$\mathrm{Ag_{51}Gd_{14}}$ phase, format $(\mathrm{Ag})_{0.785}(\mathrm{Gd})_{0.215}$
${}^{0}G^{\mathrm{Ag_{51}Gd_{14}}}_{\mathrm{Ag:Gd}} = 0.785 \times {}^{0}G^{\mathrm{fcc}}_{\mathrm{Ag}} + 0.215 \times {}^{0}G^{\mathrm{hcp}}_{\mathrm{Gd}} - 24650 + 4.358T$

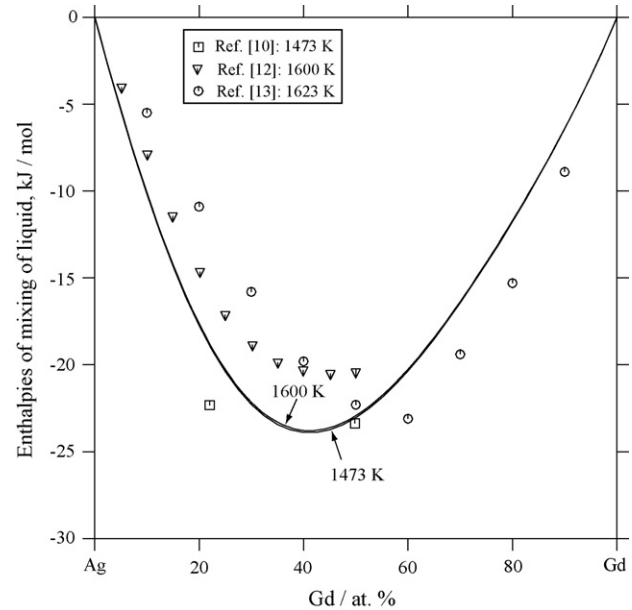

Fig. 5. The calculated enthalpies of mixing of liquid phase with reference states of pure elements of Ag (liquid) and Gd (liquid) in the Ag–Gd system compared with the experimental data [10,12,13].

Stapf and Gebhardt [17] reported three compound phases ($\mathrm{Ag_3Nd}$, $\mathrm{Ag_2Nd}$ and AgNd phases) and three eutectic reactions: L (~10 at.% Nd) ↔ (Ag) + $\mathrm{Ag_3Nd}$ at 805 °C, L (41 at.% Nd) ↔ $\mathrm{Ag_2Nd}$ + AgNd at 810 °C, L (80 at.% Nd) ↔ AgNd + αNd at 650 °C. Mulokozi et al. [18] found the $\mathrm{Ag_2Nd}$ phase to have a polymorphic transition at 620 °C.

Using the X-ray parametric method, Gschneidner et al. [6] determined the eutectic temperature (L ↔ (Ag) + $\mathrm{Ag_{51}Nd_{14}}$) to be 808 ± 3 °C and the terminal solid solubility in the fcc (Ag) solution phase to be 0.2–0.25 at.% Nd. By the X-ray powder diffraction method, Mcmasters et al. [7] reported that the stoichiometry of the Ag-rich phase in the Ag–Nd system to be $\mathrm{Ag_{51}Nd_{14}}$, which is described with an $\mathrm{Ag_{51}Gd_{14}}$-type unit cell by Bailey and Kline [8]. Furthermore, Ferro and Delfino [19] presented reasons for preferring the $\mathrm{Ag_{51}Nd_{14}}$ stoichiometry over $\mathrm{Ag_3Nd}$. Therefore, the $\mathrm{Ag_{51}Nd_{14}}$ phase was accepted in the present work.

Based on the data reported by Gschneidner et al. [6], Ferro et al. [16], Stapf and Gebhardt [17] and Mulokozi et al. [18], Gschneidner and Calderwood [20] compiled the Ag–Nd phase diagram with some modifications on the remelting reaction reported by Ferro et al. [16]. The unexplained thermal effect at 780 °C between $\mathrm{Ag_{51}Nd_{14}}$ and AgNd was ignored in the present work. The phase diagram of

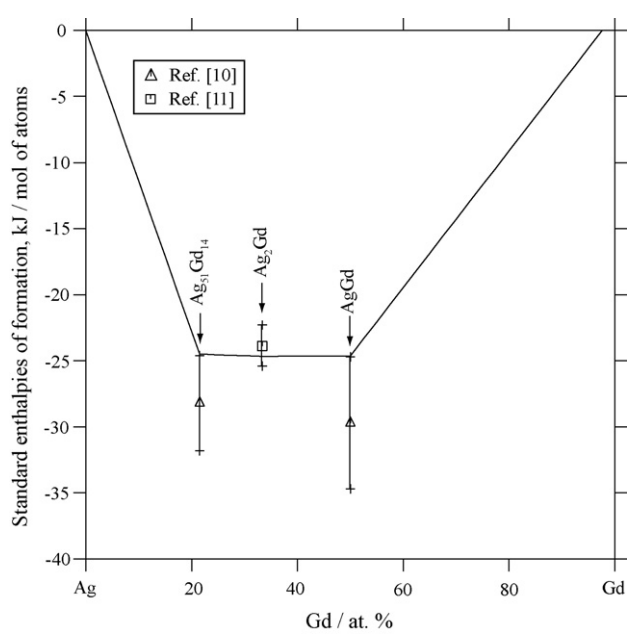

Fig. 4. The calculated standard enthalpies of formation with reference states of Ag (fcc) and αGd (hcp) in the Ag–Gd system compared with the experimental data [10,11].

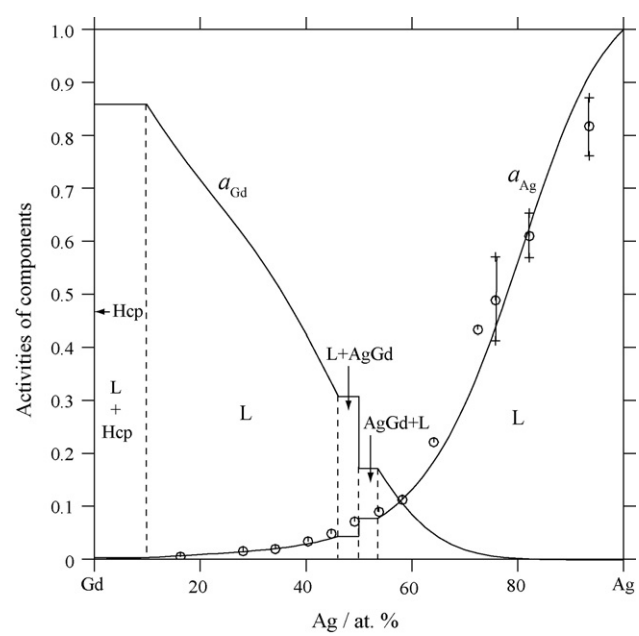

Fig. 6. The calculated activities of components at 1363 K in liquid Ag–Gd system with reference states of pure elements of Ag (liquid) and Gd (liquid) compared with the experimental data [14].

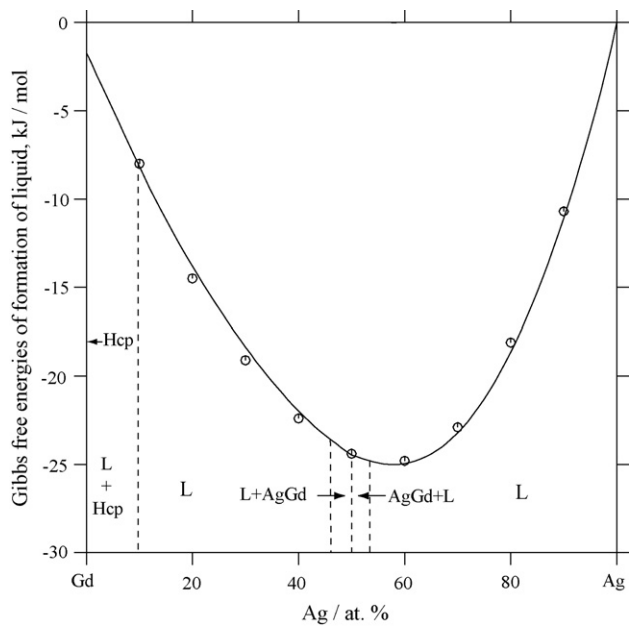

Fig. 7. The calculated Gibbs free energies of formation of liquid at 1363 K with reference states of pure elements of Ag (liquid) and Gd (liquid) compared with the experimental data [14].

Table 4
Thermodynamic parameters in the Ag–Nd system assessed in the present work.

Parameters in each phase (J/mol)
Liquid phase, format (Ag, Nd)
${}^{0}L^{\mathrm{Liq}}_{\mathrm{Ag,Nd}} = -77546 + 57.877T - 5T\ln T$
${}^{1}L^{\mathrm{Liq}}_{\mathrm{Ag,Nd}} = -23158 - 9T$
${}^{2}L^{\mathrm{Liq}}_{\mathrm{Ag,Nd}} = -2.407T$
${}^{3}L^{\mathrm{Liq}}_{\mathrm{Ag,Nd}} = 22754 - 10T$
Fcc phase, format (Ag, Nd)
${}^{0}L^{\mathrm{Fcc}}_{\mathrm{Ag,Nd}} = 9784$
${}^{1}L^{\mathrm{Fcc}}_{\mathrm{Ag,Nd}} = -41542$
Bcc phase, format (Ag, Nd)
${}^{0}L^{\mathrm{Bcc}}_{\mathrm{Ag,Nd}} = -24587 + 4.736T$
Dhcp phase, format (Ag, Nd)
${}^{0}L^{\mathrm{Dhcp}}_{\mathrm{Ag}} = {}^{0}G^{\mathrm{fcc}}_{\mathrm{Ag}} + 1000$
${}^{0}L^{\mathrm{Dhcp}}_{\mathrm{Ag,Nd}} = -9772 + 6.093T$
AgNd phase, format $(\mathrm{Ag})_{0.5}(\mathrm{Nd})_{0.5}$
${}^{0}G^{\mathrm{AgNd}}_{\mathrm{Ag:Nd}} = 0.5 \times {}^{0}G^{\mathrm{fcc}}_{\mathrm{Ag}} + 0.5 \times {}^{0}G^{\mathrm{dhcp}}_{\mathrm{Nd}} - 19500$
$\beta\mathrm{Ag_2Nd}$ phase, format $(\mathrm{Ag})_{0.667}(\mathrm{Nd})_{0.333}$
${}^{0}G^{\beta\mathrm{Ag_2Nd}}_{\mathrm{Ag:Nd}} = 0.667 \times {}^{0}G^{\mathrm{fcc}}_{\mathrm{Ag}} + 0.333 \times {}^{0}G^{\mathrm{dhcp}}_{\mathrm{Nd}} - 21000 + 2.065T$
$\alpha\mathrm{Ag_2Nd}$ phase, format $(\mathrm{Ag})_{0.667}(\mathrm{Nd})_{0.333}$
${}^{0}G^{\alpha\mathrm{Ag_2Nd}}_{\mathrm{Ag:Nd}} = 0.667 \times {}^{0}G^{\mathrm{fcc}}_{\mathrm{Ag}} + 0.333 \times {}^{0}G^{\mathrm{dhcp}}_{\mathrm{Nd}} - 21122 + 2.199T$
$\mathrm{Ag_{51}Nd_{14}}$ phase, format $(\mathrm{Ag})_{0.785}(\mathrm{Nd})_{0.215}$
${}^{0}G^{\mathrm{Ag_{51}Nd_{14}}}_{\mathrm{Ag:Nd}} = 0.785 \times {}^{0}G^{\mathrm{fcc}}_{\mathrm{Ag}} + 0.215 \times {}^{0}G^{\mathrm{dhcp}}_{\mathrm{Nd}} - 17800 + 0.118T$

the Ag–Nd system reviewed by Gschneidner and Calderwood [20] is shown in Fig. 2.

In addition, Fitzner and Kleppa [21] measured the standard enthalpies of formation of the $\mathrm{Ag_{51}Nd_{14}}$ and AgNd phases and the enthalpies of mixing of liquid phase at 1200 °C by high-temperature direct synthesis calorimetry. With the same method, Meschel and Kleppa [11] measured the standard enthalpy of formation of the $\mathrm{Ag_2Nd}$ phase. Mulokozi et al. [22] determined the heat of transformation of the $\mathrm{Ag_2Nd}$ compound to be 122 J/mol of atoms at 893 K by DSC. Ivanov and Lukashenko [14,23] measured activities of components and Gibbs free energies of formation of liquid phase at 1363 K and 1346 K by the Knudsen effusion method.

Table 3
Comparison between the calculated and experimental results in the Ag–Gd system.

Invariant reaction	Reaction type	Composition at.% Gd			Temperature (°C)	Reference
L ↔ βGd	Melting		100		1315	[5]
			100		1313	[9]
			100		1313	This work
βGd ↔ αGd	Polymorphic		100		1235	[9]
			100		1235	This work
L ↔ αGd + AgGd	Eutectic	77.5	100	50.0	790	[5]
		77.5	100	50.0	790	[9]
		75.3	100	50.0	786.3	This work
L ↔ AgGd	Congruent		50.0		1100	[5]
			50.0		1100	[9]
			50.0		1101.6	This work
L ↔ AgGd + $\mathrm{Ag_2Gd}$	Eutectic	37	50.0	33.3	825	[5]
		37	50.0	33.3	825	[9]
		34.7	50.0	33.3	826.6	This work
L + $\mathrm{Ag_{51}Gd_{14}}$ ↔ $\mathrm{Ag_2Gd}$	Peritectic	35	25.0	33.3	830	[5]
		35	21.5	33.3	830	[9]
		33.5	21.5	33.3	828.6	This work
L ↔ $\mathrm{Ag_{51}Gd_{14}}$	Congruent		25.0		975	[5]
			21.5		975	[9]
			21.5		974.7	This work
L ↔ $\mathrm{Ag_{51}Gd_{14}}$ + (Ag)	Eutectic	11	25.0	<2.0	775	[5]
			21.5	0.95	805 ± 3	[6]
		11	21.5	0.95	805 ± 3	[9]
		8.99	21.5	0.93	804.9	This work
L ↔ (Ag)	Melting		0		960.5	[5]
			0		961.9	[9]
			0		961.9	This work

3. Thermodynamic models

The information of stable solid phases and the used models in the Ag–Re (Re: Gd, Nd) binary systems are listed in Table 1.

3.1. Solution phases

In the Ag–Re (Re: Gd, Nd) binary systems, Gibbs free energies of the liquid, fcc, bcc, hcp and dhcp phases are described by the subregular solution model, as follows:

$$G_m^{\phi} = \sum_{i=\mathrm{Ag,Re}} {}^0G_i^{\phi} x_i + RT \sum_{i=\mathrm{Ag,Re}} x_i \ln x_i + \Delta^E G_m^{\phi} + \Delta^{\mathrm{mag}} G_m^{\phi}, \quad (1)$$

where ${}^0G_i^{\phi}$ is the Gibbs free energy of the pure component i in the respective reference state with the ϕ phase, which is taken from the SGTE pure element database [24]; Re means the element Gd or Nd; x_i denotes the mole fraction of the component i; R is the gas constant; T is the absolute temperature; and the term $\Delta^E G_m^{\phi}$ presents the excess Gibbs free energy, which is expressed in the Redlich–Kister polynomial [25] as:

$$\Delta^E G_m^{\phi} = x_{\mathrm{Ag}} x_{\mathrm{Re}} \sum_{m=0}^{n} {}^mL^{\phi} (x_{\mathrm{Ag}} - x_{\mathrm{Re}})^m, \quad (2)$$

$${}^iL_{\mathrm{Ag,Re}}^{\phi} = a + bT + cT \ln T, \quad (3)$$

Table 5
Comparison between the calculated and experimental results in the Ag–Nd system.

Invariant reaction	Reaction type	Composition at.% Nd			Temperature (°C)	Reference
L ↔ βNd	Melting		100		1017	[16]
			100		1017	[17]
			100		1021	[20]
			100		1021	This work
βNd ↔ αNd	Polymorphic		100		847	[16]
			100		863	[20]
			100		863	This work
βNd ↔ L + αNd	Remelting	~96		~99	725	[16]
					741	[20]
		96.1	85.4	99.2	742.8	This work
L ↔ αNd + AgNd	Eutectic	79.5		50.0	630 ± 5	[16]
		80	100		650	[17]
		80		50.0	640	[20]
		79.9	99.1	50.0	642.4	This work
L ↔ AgNd	Congruent		50.0		960 ± 10	[16]
			50.0		945	[17]
			50.0		945	[18]
			50.0		952	[20]
			50.0		950.2	This work
L ↔ AgNd + βAg_2Nd	Eutectic	40.8	50.0	33.3	810 ± 5	[16]
		41	50.0	33.3	810	[17]
			50.0	33.3	810	[18]
		40.8	50.0	33.3	810	[20]
		37.8	50.0	33.3	811.8	This work
L + $Ag_{51}Nd_{14}$ ↔ βAg_2Nd	Peritectic		21.5	33.3	831 ± 5	[16]
		35	25.0	33.3	820	[17]
			25.0	33.3	820	[18]
				33.3	825	[20]
		35.6	21.5	33.3	823.9	This work
βAg_2Nd ↔ αAg_2Nd	Polymorphic		33.3		650 ± 10	[16]
					620	[17]
					620	[18]
			33.3		635.0	[20]
			33.3		635.0	This work
L ↔ $Ag_{51}Nd_{14}$	Congruent		21.5		1025 ± 10	[16]
			25		955	[17]
			25		955	[18]
					1025	[20]
			21.5		1024.5	This work
L ↔ $Ag_{51}Nd_{14}$ + (Ag)	Eutectic	10.5	21.5	~0.2	803 ± 5	[16]
		10.0	25.0	0	805	[17]
			21.5		808 ± 3	[6]
		10.5		~0.2	806	[20]
		8.4	21.5	0.31	802.3	This work
L ↔ (Ag)	Melting		0		960	[16]
			0		960.5	[17]
			0		961.9	[20]
			0		961.9	This work

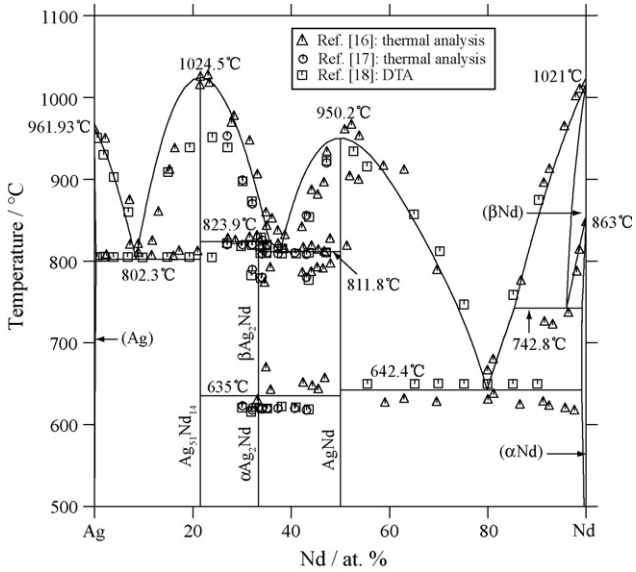

Fig. 8. The calculated phase diagram of the Ag–Nd system compared with the experimental data [16–18].

where ${}^{i}L^{\phi}_{\mathrm{Ag,Re}}$ is the binary interaction parameter, and the coefficients of a, b and c are constants to be evaluated on the basis of available experimental data.

$\Delta^{\mathrm{mag}}G^{\phi}_{m}$ is the magnetic contribution to the Gibbs free energy, which is described by the following equation [26]:

$$\Delta^{\mathrm{mag}}G^{\phi}_{m} = RT\ \ln(\beta+1)f(\tau) \tag{4}$$

where $\tau = T/T^{\phi}_{\mathrm{C}}$, T^{ϕ}_{C} is the Curie temperature of solution for ferromagnetic ordering and β is the Bohr magneton number, the function $f(\tau)$ is formulated by the polynomial of the normalized temperature, as follows:

$$f(\tau) = 1 - \left[\frac{79\tau^{-1}}{140s} + \frac{474}{497}\left(\frac{1}{s}-1\right)\left(\frac{\tau^{3}}{6}+\frac{\tau^{9}}{135}+\frac{\tau^{15}}{600}\right)\right]/D \tag{5}$$

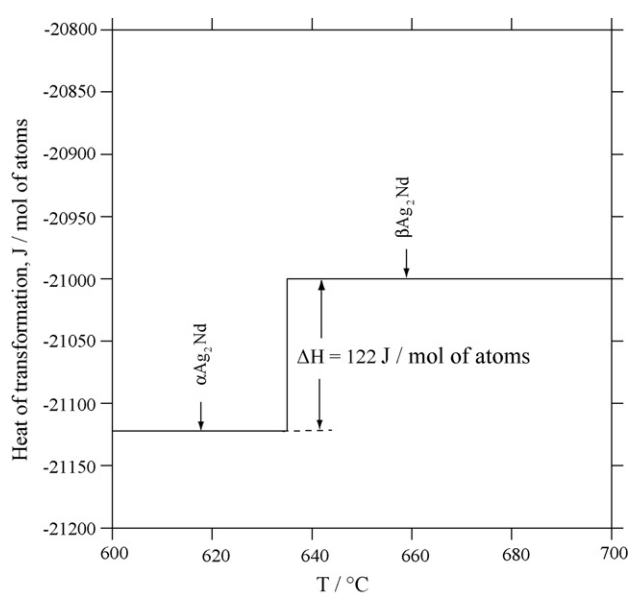

Fig. 9. The calculated heat of transformation of the Ag_2Nd compound at 600–700 °C.

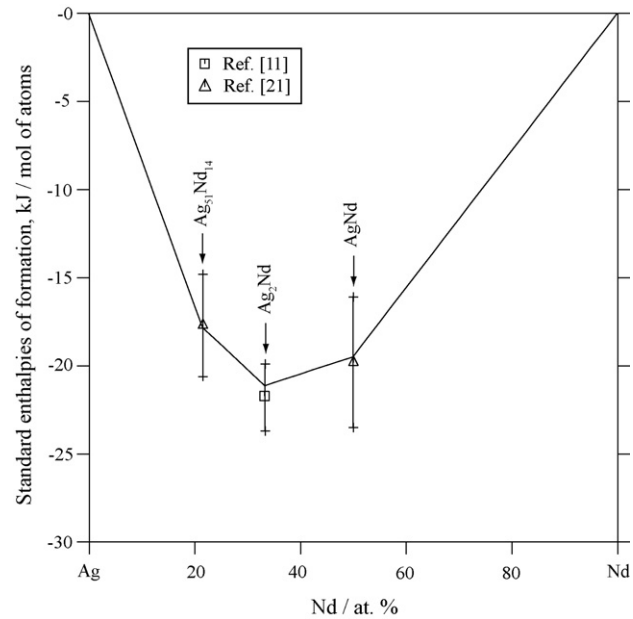

Fig. 10. The calculated standard enthalpies of formation with reference states of Ag (fcc) and αNd (dhcp) in the Ag–Nd system compared with the experimental data [11,21].

$$f(\tau) = \frac{-(\tau^{-5}/10 + \tau^{-15}/315 + \tau^{-25}/1500)}{D} \quad \text{for } \tau > 1 \tag{6}$$

$$D = \frac{518}{1125} + \frac{11692}{15975}\left(\frac{1}{s}-1\right) \tag{7}$$

where s depends on the structure, 0.40 for bcc structure and 0.28 for others.

Since the hcp phase in Ag–Gd system is magnetic phase, thus the magnetic contribution ($\Delta^{\mathrm{mag}}G^{\mathrm{hcp}}_{m}$) to the Gibbs free energy was considered in the present work.

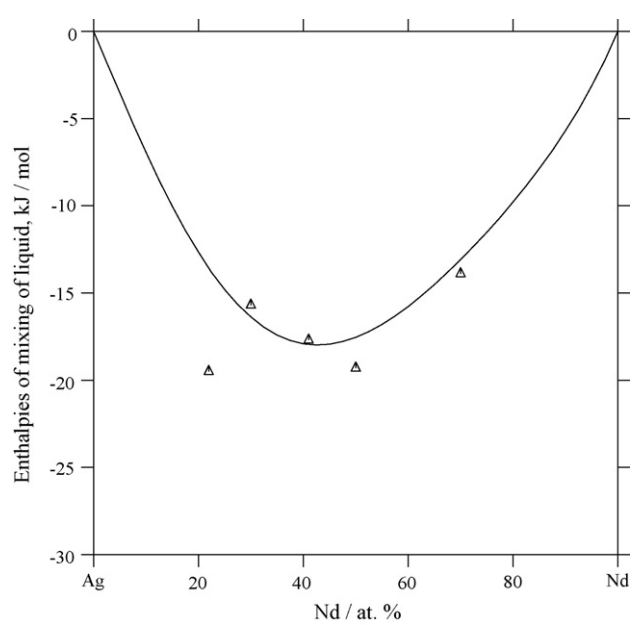

Fig. 11. The calculated enthalpies of mixing of liquid phase at 1200 °C with reference states of pure elements of Ag (liquid) and Nd (liquid) in the Ag–Nd system compared with the experimental data [21].

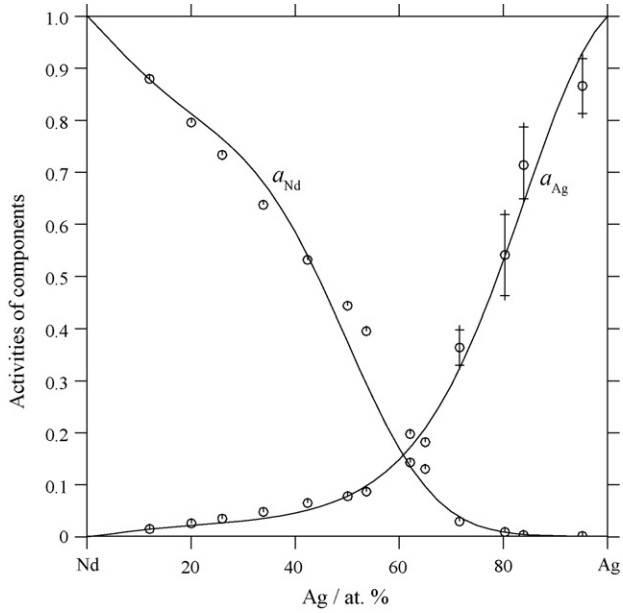

Fig. 12. The calculated activities of components at 1346 K in liquid Ag–Nd system with reference states of pure elements of Ag (liquid) and Nd (liquid) compared with the experimental data [23].

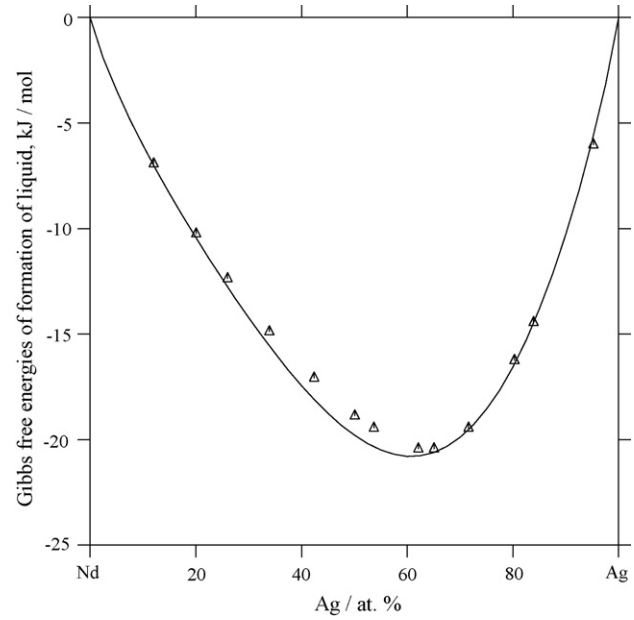

Fig. 14. The calculated Gibbs free energies of formation of liquid at 1346 K with reference states of pure elements of Ag (liquid) and Nd (liquid) compared with the experimental data [23].

3.2. Stoichiometric intermetallic compounds

The intermetallic compounds of the $Ag_{51}Gd_{14}$, Ag_2Gd, AgGd, $Ag_{51}Nd_{14}$, αAg_2Nd, βAg_2Nd and AgNd phases in the Ag–Re (Re: Gd and Nd) binary systems are all treated as stoichiometric phases, because no reliable experimental data about the homogeneity range of these compounds are available. The Gibbs free energy per mole of formula unit Ag_pRe_q can be expressed, as follows:

$$G^{Ag_pRe_q}_{Ag:Re} = p^0G^{SER}_{Ag} + q^0G^{SER}_{Re} + \Delta^0G^{Ag_pRe_q}_f + \Delta^{mag}G^{Ag_pRe_q}_m, \quad (8)$$

where the $\Delta^0G^{Ag_pRe_q}_f$ indicates the standard Gibbs free energy of formation of the stoichiometric compound from pure elements, which is described as:

$$\Delta^0G^{Ag_pRe_q}_f = a' + b'T, \quad (9)$$

where the parameters of a' and b' are evaluated in the present work. $\Delta^{mag}G^{Ag_pRe_q}_m$ is magnetic contribution to the Gibbs free energy, which is expressed in Eqs. (4)–(7), where β and T^{ϕ}_C of each intermetallic compound are taken from literature [15].

Since the AgGd intermetallic compound is magnetic phase, thus the magnetic contribution ($\Delta^{mag}G^{AgGd}_m$) to the Gibbs free energy was considered in the present work.

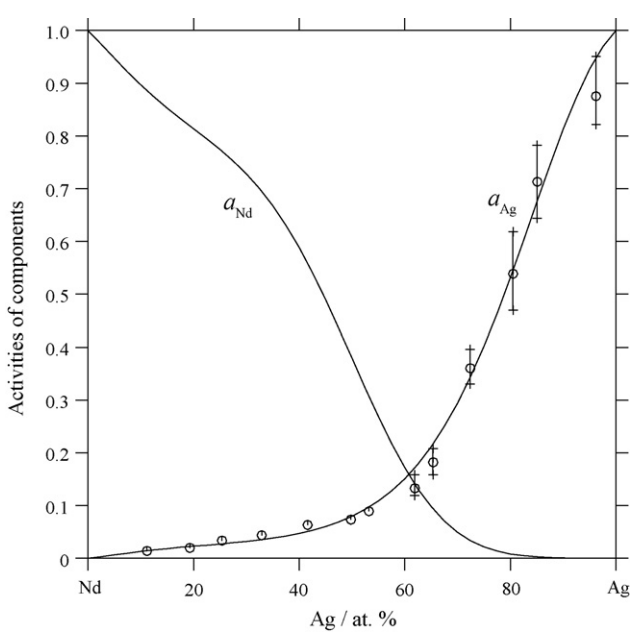

Fig. 13. The calculated activities of components at 1363 K in liquid Ag–Nd system with reference states of pure elements of Ag (liquid) and Nd (liquid) compared with the experimental data [14].

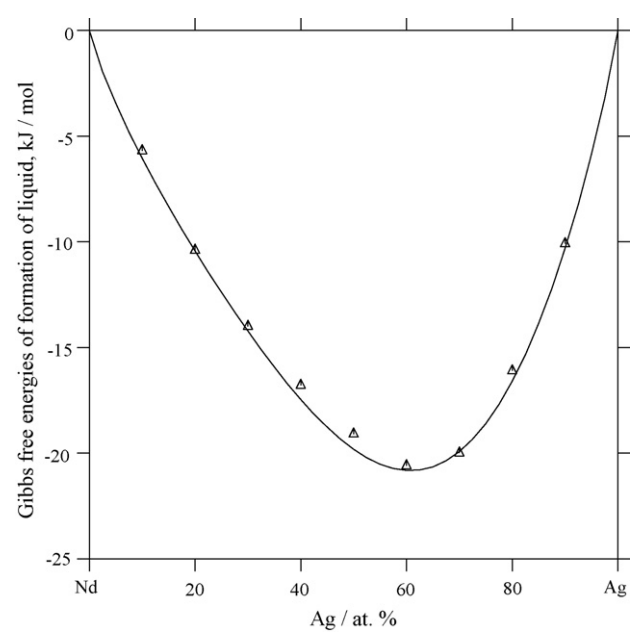

Fig. 15. The calculated Gibbs free energies of formation of liquid at 1363 K with reference states of pure elements of Ag (liquid) and Nd (liquid) compared with the experimental data [14].

4. Optimized results and discussion

The optimization of the thermodynamic parameters was carried out by using the PARROT program in the Thermo-Calc software [27], which can handle various kinds of experimental data. The experimental data of the phase diagram and thermodynamic properties were used as input to the program. Each piece of selected information was given a certain weight based on the importance of data, and changed by trial and error during the assessment, until most of the selected experimental information was reproduced within the expected uncertainty limits.

4.1. The Ag–Gd system

The calculated Ag–Gd phase diagram compared with the experimental data is shown in Fig. 3. It is seen that the calculated results are in reasonable agreement with the experimental data [5,6]. A set of complete self-consistent thermodynamic parameters describing the Gibbs free energy of each phase in the Ag–Gd system are given in Table 2, and the calculated compositions and temperatures for the invariant reactions compared with the selected experimental data are listed in Table 3.

The calculated standard enthalpies of formation with reference states of the Ag (fcc) and αGd (hcp) phases compared with the experimental data are presented in Fig. 4, and a comparison of enthalpies of mixing of liquid phase between the calculated and experimental results is shown in Fig. 5. The calculated activities with experimental data in the liquid binary alloys at 1363 K with reference states of pure elements of Ag (liquid) and Gd (liquid) are illustrated in Fig. 6. Fig. 7 shows the calculated Gibbs free energies of formation of liquid phase with experimental data at 1363 K with reference states of pure elements of Ag (liquid) and Gd (liquid). It is seen that a good agreement on the thermodynamic properties is obtained between the calculated results and the experimental data.

4.2. The Ag–Nd system

The calculated Ag–Nd phase diagram compared with the experimental data is shown in Fig. 8. It is seen that the calculated results are in reasonable agreement with the experimental data [16–18]. A set of complete self-consistent thermodynamic parameters describing the Gibbs free energy of each phase in the Ag–Nd system are given in Table 4, and the calculated compositions and temperatures for the invariant reactions compared with the selected experimental data are listed in Table 5.

Fig. 9 shows the calculated heat of transformation of the Ag_2Nd compound at 600–700 °C, which is in agreement with the experimental data [22]. The calculated standard enthalpies of formation with reference states of the Ag (fcc) and αNd (dhcp) phases compared with the experimental data are presented in Fig. 10, and a comparison of enthalpies of mixing of liquid phase at 1200 °C between the calculated and experimental results is shown in Fig. 11. The calculated activities with experimental data in the liquid binary alloys at 1346 and 1363 K with reference states of pure elements of Ag (liquid) and Nd (liquid) are illustrated in Figs. 12 and 13. Figs. 14 and 15 show the calculated Gibbs free energies of formation of liquid phase with experimental data at 1346 and 1363 K with reference states of pure elements of Ag (liquid) and Nd (liquid). It is seen that a good agreement on the thermodynamic properties is obtained between the calculated results and the experimental data.

5. Conclusions

The phase diagram in the Ag–Re (Re: Gd, Nd) binary systems was critically assessed by considering the experimental data including phase equilibria and thermodynamic data. A consistent set of thermodynamic parameters has been optimized and most of the experimental information can be satisfactorily reproduced on the basis of the optimized thermodynamic parameters.

Acknowledgements

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